

Automatic perturbation kernels for biological applications of Approximate Bayesian Computation sequential Monte Carlo with random forests

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Abstract—We recently introduced Approximate Bayesian Computation sequential Monte Carlo with random forests (ABC-SMC-RF), a novel likelihood-free inference method that combines the robustness of random forests and the efficiency of the Approximate Bayesian Computation sequential Monte Carlo (ABC-SMC) framework. In this work, we examine the selection of parameter perturbation kernels and their impact on ABC-SMC-RF’s results. We evaluate the performance of an automatic and adaptive perturbation scheme, previously developed for ABC-SMC. Across examples drawn from different realistic biological contexts, we show that (i) ABC-SMC-RF produces precise and accurate parameter inferences, without requiring extensive hyper-parameter tuning, and (ii) the adaptive perturbations yield optimal performance in ABC-SMC-RF, further reducing practical barriers to its adoption in many settings. We expect ABC-SMC-RF, armed with the automatic perturbation scheme, to be well positioned for supporting different scientific problems that require robust, efficient and convenient parameter estimation.

I. INTRODUCTION

PARAMETER inference is a crucial component of mathematical modeling, enabling researchers to estimate variables of interest from data, evaluate whether a proposed model can explain observations, and analyze scientific phenomena in intimate details. In the Bayesian framework, estimating the parameters θ involves computing their posterior distribution

$$\pi(\theta|z^{\text{obs}}) \propto \mathcal{L}(\theta|z^{\text{obs}}) \cdot \pi(\theta)$$

where the prior distribution $\pi(\theta)$ reflects *a priori* assumptions for where the parameters reside, and the likelihood $\mathcal{L}(\theta|z^{\text{obs}})$ determines the probability of extracting the observed data z^{obs} from the proposed model, given parameter values θ .

In cases where the likelihood function is intractable or numerically challenging, Approximate Bayesian Computation (ABC) has been proposed as an alternative method [1]. ABC variations, including the rejection method [1], Monte Carlo Markov chain (MCMC) [2], and sequential Monte Carlo (SMC) [3], seek the parameter distributions that minimize the distance between statistics from model simulations and those of the observed data. Despite significant success across

diverse scientific fields, implementing ABC to estimate parameters accurately and efficiently in practice relies on intensive hyper-parameter calibration. First, ABC requires a metric to define the distance between observed and simulated data. Weighting the metric to reflect both the scale and relevance of different statistics can be challenging. Second, choosing tolerance thresholds to accept or reject proposed parameters based on their distances to data requires intuition or experimentation to balance between efficiency and accuracy. Finally and most importantly, ABC’s convergence to the likelihood-based posterior is only guaranteed if the statistics are sufficient. For complex models, low-dimensional sufficient features may not be available, thus researchers may risk distorting the estimations by including additional uninformative statistics.

II. INTEGRATING RANDOM FORESTS IN ABC

Recent work has sought to alleviate the practical shortcomings of traditional ABC approaches by implementing random forests in the context of Bayesian inference. ABC random forest (ABC-RF) [4], developed by Raynal et al., first builds $\mathcal{N}_t$ random trees from a reference table $\mathcal{T}$ of size $\mathcal{N}_p$. Each particle $p = 1, \dots, \mathcal{N}_p$ in $\mathcal{T}$ consists of a parameter $\theta^{(p)}$ sampled from $\pi(\theta)$ and the corresponding simulated statistics $z^{(p)}$. Each random tree $t = 1, \dots, \mathcal{N}_t$ is initiated from a root node, bootstrapped from $\mathcal{T}$. Each node R_* in the tree selects one statistic z_* and a threshold k_* , then splits the particles in R_* into two nodes $R_{*1} = \{p \in R_* : z_*^{(p)} < k_*\}$ and $R_{*2} = \{p \in R_* : z_*^{(p)} \geq k_*\}$. The choice of z_* and k_* are selected such that the L^2 -loss between R_{*1} and R_{*2} is minimal. To optimize forest diversity, the set of potential $\{z_*\}$ in consideration for splitting each node is limited to a randomly selected subset among all available statistics. The decision tree is finished when each leaf contains fewer than 5 particles, or all particles in each leaf have the same statistics.

Once all $\mathcal{N}_t$ trees are constructed, ABC-RF inputs the statistics z^{obs} observed from data into the root node of each tree t , follows the predetermined splitting rules to arrive

at a leaf, then extracts the set of particles $L_t(z^{\text{obs}})$ in the selected leaf. Raynal et al. argued that the approximate posterior distribution is then the density of all parameter values $\{\theta^{(p)}, p = 1, \dots, \mathcal{N}_p\}$, where the weight for particle p is

$$w^{(p)} = \frac{1}{\mathcal{N}_t} \sum_{t=1}^{\mathcal{N}_t} \frac{n_t^{(p)} \cdot \mathbf{1}(p \in L_t(z^{\text{obs}}))}{\sum_{p'} n_t^{(p')} \cdot \mathbf{1}(p' \in L_t(z^{\text{obs}}))}$$

with $n_t^{(p)}$ being the number of times that particle p is duplicated in the bootstrapped root node for tree t , and $\mathbf{1}(p \in L)$ is a function that equals 1 if particle $p \in L$ and 0 otherwise.

For multivariate models, Čevid et al. developed distributional random forests (DRF) [5]. In addition to implementing L^2 -loss to determine node splitting, the authors also constructed a second metric based on maximal mean discrepancy (MMD), which considers the difference in distribution between particles in the child nodes. DRF also employs different strategies to ensure forest diversity, compared to ABC-RF. Each root node in DRF is subsampled, instead of bootstrapped, from the training set. By default, DRF further divides these particles into disjoint halves, where the tree is constructed from one particle subset and the weights are determined from the other. Finally, the total number of randomly selected statistics considered to split each node is itself randomly sampled from a Poisson distribution, instead of being fixed as in ABC-RF.

III. APPROXIMATE BAYESIAN COMPUTATION SEQUENTIAL MONTE CARLO WITH RANDOM FORESTS

ABC algorithms with random forests offer several remarkable advantages compared to the traditional ABC approaches. First, ABC-RF and DRF do not require metrics to compare data and simulated statistics, or tolerance thresholds to determine if a given particle should be accepted or rejected, the choices of both can impact the posterior approximation significantly. Furthermore, Raynal et al. showed that ABC-RF is considerably insensitive to noise [4]. Therefore, for high-dimensional problems, the implementation of ABC with random forests can accommodate a large selection of statistics, with less risk of introducing noise in the parameter estimation.

However, the construction of random forests poses practical limitations on their application for complicated problems. The runtime and storage for a random forest of $\mathcal{N}_t$ trees built on a training set of size $\mathcal{N}_p$ scale with $\mathcal{O}(\mathcal{N}_t \cdot \mathcal{N}_p \cdot \log \mathcal{N}_p)$. For complex models and datasets where larger forests and more particles are required to improve the parameter distribution, the computational requirements can be inhibitive.

We recently introduced ABC sequential Monte Carlo with random forests (ABC-SMC-RF) [6]. There are two versions of ABC-SMC-RF, constructed for different classes of scientific problems. The multivariate ABC-SMC-RF algorithm (Alg. 1) was designed for problems where model parameters are expected to be highly dependent. Its framework is based on ABC sequential Monte Carlo (ABC-SMC), where the posterior density is iteratively refined until it converges to the true distribution [3]. In the first iteration, particles are simulated with parameters sampled from the prior distribution.

Unlike ABC-SMC, which then accepts or rejects particles by comparing their distances to the data statistics with prescribed tolerance thresholds, ABC-SMC-RF assembles a reference table containing all simulated particles. It then performs DRF on this training set, and computes particle weights based on given statistics from data. In the next iterations, ABC-SMC-RF samples parameter sets from the previous run's weighted density, then perturbs the parameters with a Markov kernel and simulates corresponding statistics. DRF is performed again on the new training set, and the resulting weights are recalibrated with ABC-SMC's formula to account for the kernel's impact (Step 17, Alg. 1). The weighted density from the final iteration then forms ABC-SMC-RF's approximation for the posterior.

The second ABC-SMC-RF algorithm is applicable for models either consisting of a single parameter or whose parameters are relatively independent (Alg. 2). Instead of running DRF for all parameters simultaneously in each iteration, it performs ABC-RF on a training set where the particles are reduced to each parameter. The parameter's weights are similarly adjusted with ABC-SMC's recalibration formula (Step 21, Alg. 2). In iterations $2, \dots, I$, new particles are sampled marginally, where their values for a given parameter are drawn from its weighted density in the previous iteration.

ABC-SMC-RF offers three practical advantages, compared to ABC-RF and DRF. First, the random forest-related computational cost for I iterations where each iteration i constructs $\mathcal{N}_t$ trees from a training set of $\mathcal{N}_p^{(i)}$ particles is $\mathcal{O}(\mathcal{N}_t \cdot \sum_{i=1}^I \mathcal{N}_p^{(i)} \cdot \log \mathcal{N}_p^{(i)})$. Compared to single-iteration ABC-(D)RF based on the same total particle count (i.e., $\mathcal{N}_p = \sum_{i=1}^I \mathcal{N}_p^{(i)}$) with complexity $\mathcal{O}(\mathcal{N}_t \cdot \mathcal{N}_p \cdot \log \mathcal{N}_p)$, ABC-SMC-RF can lead to significant reductions in runtime and storage. Second, as the posterior density is successively refined, the proposed particles are increasingly informed. Therefore, the weighted density from ABC-SMC-RF is expected to converge more quickly to the true posterior distribution, thus improving accuracy and efficiency. Finally, the iterative nature of ABC-SMC's framework allows for monitoring the convergence rate of the posterior approximation. In practice, if successive parameter densities are significantly different, indicating that the process has not converged, the user may extend a completed ABC-SMC-RF run with additional iterations, toward improving the final parameter estimations.

IV. PERTURBATION KERNELS FOR ABC-SMC-RF

Despite ABC-SMC-RF's real-world advantages over ABC-RF and DRF, its applications require the users to define the Markov kernels for perturbing iteratively sampled parameters (Step 7, Alg. 1 and Step 8, Alg. 2). Parameter perturbation is an essential part of the ABC-SMC framework, in order to encourage the particles to explore the parameter space and simultaneously maintain particle diversity [7]. The choice of kernels must balance between being broad enough to enable sufficient exploration and being adequately narrow so that the parameter density can converge. Therefore, the optimal selection of perturbation kernels often requires appropriate tuning to the scale and structure of the model parameters.

Algorithm 1 ABC-SMC-RF for multiple parameters

```

1: for iteration  $i = 1, \dots, I$  do
2:   for particle  $p = 1, \dots, \mathcal{N}_p^{(i)}$  do
3:     if  $i = 1$  then
4:       Sample  $\theta_1^{(p)}$  from prior distribution  $\pi(\theta)$ 
5:     else
6:       Sample  $\theta^*$  from  $\left\{ \left( \theta_{i-1}^{(p')}, w_{i-1}^{(p')} \right) \right\}_{p'=1, \dots, \mathcal{N}_p^{(i-1)}}$ 
7:       Generate  $\theta_i^{(p)} \sim K_i(\theta|\theta^*)$ 
8:       If  $\pi(\theta_i^{(p)}) = 0$ , then return to Step 6
9:     end if
10:    Simulate statistics  $z_i^{(p)}$  with parameters  $\theta_i^{(p)}$ 
11:  end for
12:  Form training set  $\mathcal{T}_i = \left\{ \left( \theta_i^{(p)}, z_i^{(p)} \right) \right\}_{p=1, \dots, \mathcal{N}_p^{(i)}}$ 
13:  Perform DRF with reference table  $\mathcal{T}_i$  and  $\mathcal{N}_t$  trees to
  compute particle weights  $w_i^{*(1)}, \dots, w_i^{*(\mathcal{N}_p^{(i)})}$ 
14:  if  $i = 1$  then
15:     $w_i^{(p)} = w_i^{*(p)}$  for  $p = 1, \dots, \mathcal{N}_p^{(i)}$ 
16:  else
17:     $w_i^{(p)} = w_i^{*(p)} \frac{\pi(\theta_i^{(p)})}{\sum_{p'=1}^{\mathcal{N}_p^{(i-1)}} w_{i-1}^{(p')} \cdot K_i(\theta_i^{(p)}|\theta_{i-1}^{(p')})}$ 
18:  end if
19:  Normalize the particle weights  $\left\{ w_i^{(p)} \right\}_{p=1, \dots, \mathcal{N}_p^{(i)}}$ 
20: end for

```

Beaumont et al. introduced an adaptive parameter perturbation scheme for the ABC-SMC algorithm [8]. In this process, the values for parameter j in iteration i , sampled from particles in iteration $i - 1$, are perturbed with Gaussian noise

$$K_{j,i}(\theta|\theta^*) = \theta^* + \mathcal{N}(0, 2 \cdot \sigma_{j,i-1}^2) \quad (1)$$

where $\sigma_{j,i-1}^2$ is the empirical variance in iteration $i - 1$.

Eq. (1) provides an automatic and adaptive perturbation scheme that is constantly updated to reflect the uncertainty in current iteration's particles. Beaumont et al. proved that $2 \cdot \sigma_{j,i-1}^2$ is the optimal perturbation scale with minimized Kullback divergence. The perturbation scheme (1) has since become the default choice for most ABC-SMC applications.

V. ABC-SMC-RF APPLICATIONS IN BIOLOGY

We now examine ABC-SMC-RF's performance in realistic biological contexts, compared to both traditional ABC approaches and previous ABC algorithms with random forests. In addition, for ABC-SMC-RF, we also investigate the effectiveness of the adaptive perturbation scheme (1), hereby named "Beaumont perturbations", against heuristic strategies. To ensure fair comparison, all algorithms are configured to employ roughly the same total particle counts in each problem.

A. Stochastic Lotka-Volterra model

We consider the stochastic Lotka-Volterra model [9], [10]:

$$\begin{aligned}
 x &\rightarrow x + 1, \text{ with propensity } \alpha \cdot x \\
 x &\rightarrow x - 1, \text{ with propensity } (\beta/1000) \cdot x \cdot y
 \end{aligned}$$

Algorithm 2 ABC-SMC-RF for single parameters

```

1: for iteration  $i = 1, \dots, I$  do
2:   for particle  $p = 1, \dots, \mathcal{N}_p^{(i)}$  do
3:     if  $i = 1$  then
4:       Sample  $\theta_1^{(p)}$  from prior distribution  $\pi(\theta)$ 
5:     else
6:       for parameter  $j = 1, \dots, J$  do
7:         Sample  $\theta_j^* \sim \left\{ \left( \theta_{j,i-1}^{(p')}, w_{j,i-1}^{(p')} \right) \right\}_{p'=1, \dots, \mathcal{N}_p^{(i-1)}}$ 
8:         Generate  $\theta_{j,i}^{(p)} \sim K_{j,i}(\theta|\theta_j^*)$ 
9:         If  $\pi_j(\theta_{j,i}^{(p)}) = 0$ , then return to Step 7
10:       end for
11:       Assemble  $\theta_i^{(p)} = \left( \theta_{1,i}^{(p)}, \dots, \theta_{J,i}^{(p)} \right)$ 
12:     end if
13:     Simulate statistics  $z_i^{(p)}$  with parameters  $\theta_i^{(p)}$ 
14:   end for
15:   for parameter  $j = 1, \dots, J$  do
16:     Form training set  $\mathcal{T}_{j,i} = \left\{ \left( \theta_{j,i}^{(p)}, z_i^{(p)} \right) \right\}_{p=1, \dots, \mathcal{N}_p^{(i)}}$ 
17:     Perform ABC-RF with reference table  $\mathcal{T}_{j,i}$  and  $\mathcal{N}_t$ 
     trees to compute particle weights  $w_{j,i}^{*(1)}, \dots, w_{j,i}^{*(\mathcal{N}_p^{(i)})}$ 
18:     if  $i = 1$  then
19:        $w_{j,i}^{(p)} = w_{j,i}^{*(p)}$  for  $p = 1, \dots, \mathcal{N}_p^{(i)}$ 
20:     else
21:        $w_{j,i}^{(p)} = w_{j,i}^{*(p)} \frac{\pi_j(\theta_{j,i}^{(p)})}{\sum_{p'=1}^{\mathcal{N}_p^{(i-1)}} w_{j,i-1}^{(p')} \cdot K_{j,i}(\theta_{j,i}^{(p)}|\theta_{j,i-1}^{(p')})}$ 
22:     end if
23:     Normalize the particle weights  $\left\{ w_{j,i}^{(p)} \right\}_{p=1, \dots, \mathcal{N}_p^{(i)}}$ 
24:   end for
25: end for

```

$$y \rightarrow y + 1, \text{ with propensity } (\delta/1000) \cdot x \cdot y$$

$$y \rightarrow y - 1, \text{ with propensity } \gamma \cdot y$$

where x and y are the numbers of prey and predators, respectively. The propensities define the inverse means of the exponentially distributed wait times for each next event to occur. The entities α and β parametrize the events of prey multiplying and being hunted by predators, respectively. Likewise, δ and γ are the parameters of predator growing and death. The observed data consists of population sizes $(x_n^{\text{obs}}, y_n^{\text{obs}})$ at times $t_n = 1, \dots, 14$, created with the Stochastic Simulation Algorithm (SSA) [11] starting from initial population sizes $(x(0), y(0)) = (700, 300)$ with true parameter values $\alpha = \beta = \delta = \gamma = 1$. We seek to infer parameters $\theta = \{\alpha, \beta, \delta, \gamma\}$ from the observed data with prior distributions

$$\pi(\alpha) = \pi(\beta) = \pi(\delta) = \pi(\gamma) = \text{Uniform}(0, 2)$$

We first implement ABC-rejection [1], where a total of 12500 particles are sampled. For each particle p , the population trajectory $(x(t), y(t))$ is simulated with the proposed parameter values, then its distance to the data is defined as

$$d(z, z^{\text{obs}}) = \sum_{n=1}^{14} \left((x(t_n) - x_n^{\text{obs}})^2 + (y(t_n) - y_n^{\text{obs}})^2 \right) \quad (2)$$

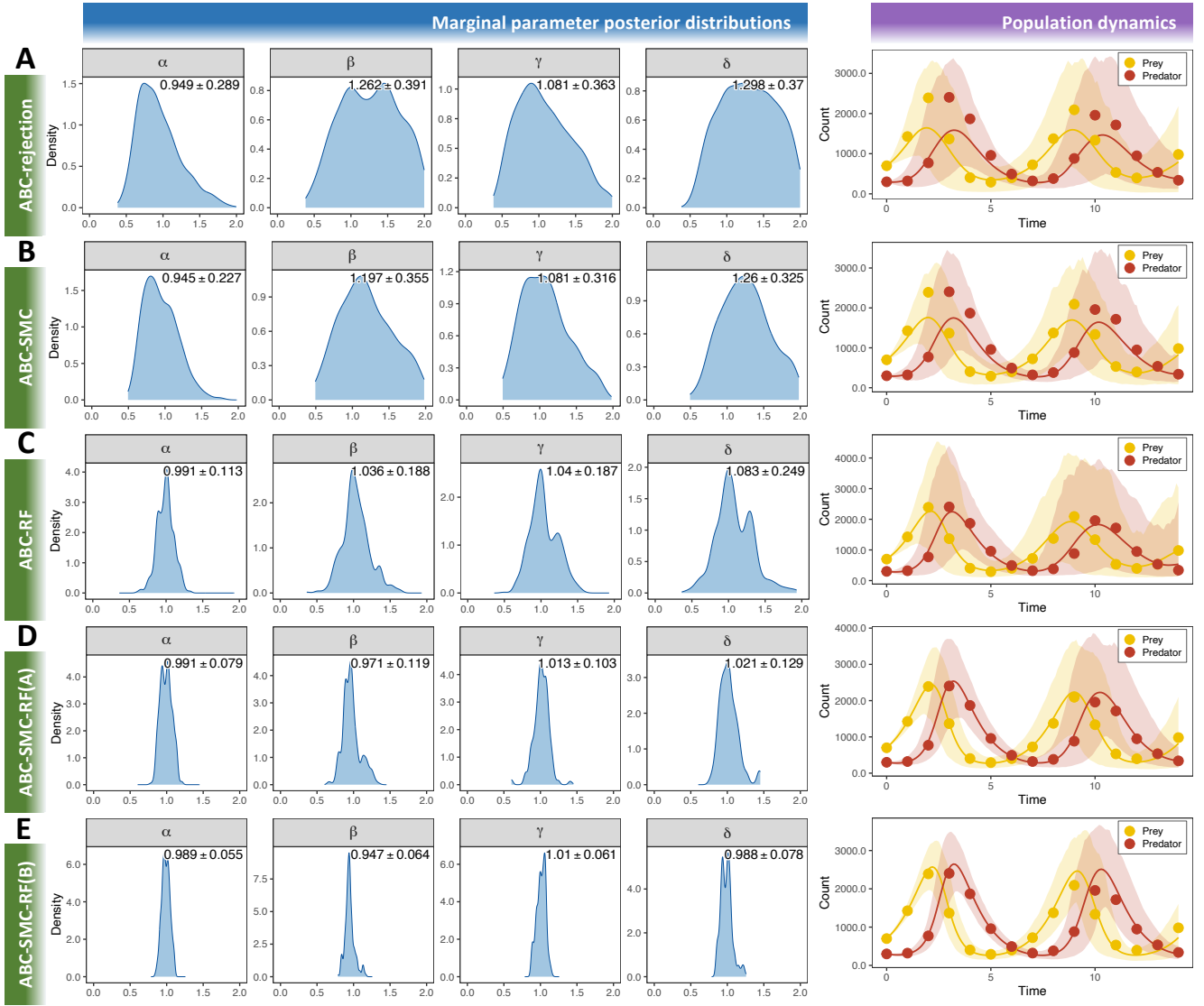


Fig. 1. Comparison of marginal posteriors (left) and corresponding simulated population dynamics (right) from ABC-rejection (A), ABC-SMC (B), ABC-RF (C), ABC-SMC-RF(A) (D) and ABC-SMC-RF(B) (E) for the stochastic Lotka-Volterra model (details in Section V-A). Each algorithm is configured such that roughly the same total number of particles (≈ 12500) are drawn and simulated. Numbers in each parameter's density plot indicate the mean $\pm$ standard deviation from the approximated marginal posterior distribution. The population dynamic plots compare observed data (dots) against the temporal means (lines) and 95% confidence intervals (shaded areas) of prey and predator population sizes, simulated with parameters sampled from the posterior distributions.

Finally, we extract 500 particles with the lowest distances (i.e., particles in the top 4% percentile are accepted), and their values of θ constitute the posterior density. We observe that the resulting posterior distribution has moved toward the true parameter values of $\theta = (1, 1, 1, 1)$ (Fig. 1A). However, each parameter's standard deviation remains large, and population trajectories simulated from the parameter posterior distributions are relatively inaccurate compared to the observed data, both indicating that the results can be improved by increasing the particle count and lowering the tolerance threshold.

We then execute ABC-SMC with the R library *EasyABC* [12]. The ABC-SMC routine is designed to complete 4 iterations, where each iteration uses the same statistic distance

(2) with decreasing tolerance thresholds to accept or reject particles ($\epsilon_1 = 0.1$, $\epsilon_2 = 0.03$, $\epsilon_3 = 0.01$, $\epsilon_4 = 0.005$), toward producing 200 particles in iteration 4. The program employs 12553 particles in total. The posterior distributions are tighter compared to ABC-rejection, and the simulated population dynamics are closer to the data (Fig. 1B). This is a direct result of the ABC-SMC framework, where the posterior density is iteratively refined so that the accepted particles in the final iteration can better capture the patterns in the data statistics, even for similar total particle counts to ABC-rejection.

Next, we employ ABC-RF from [4]. The training set consists of 12500 particles, sampled from the prior distribution. ABC-RF does not require a distance function to compare

data and simulated statistics (e.g., Eq. (2)) or configuring the tolerance thresholds. Instead, it employs the simulated population sizes $(x(t_n), y(t_n))_{n=1, \dots, 14}$ directly to construct the random forest. Despite no model-specific fine-tuning, the posterior densities, weighted for the observed data $(x_n^{\text{obs}}, y_n^{\text{obs}})$, are significantly narrower compared to ABC-rejection or ABC-SMC, with means within 10% of the true parameter values (Fig. 1C). Additionally, the averages of population dynamics simulated from these posterior densities closely resemble the data. These results underline the power of ABC approaches with random forests: they require minimal hyper-parameters, yet are able to produce highly accurate parameter estimations.

We also analyze the performance of ABC-SMC-RF for single parameters (Alg. 2; [6]) with 5 iterations, each with 2500 particles. Similar to ABC-RF, all simulated and observed population sizes are utilized to construct random forests and compute particle weights. We first implement heuristic Gaussian perturbation kernels

$$K_{j,i}(\theta|\theta^*) = \theta^* + \mathcal{N}(0, \sigma^2 = 0.2^2)$$

for each iteration i and parameter $j = \alpha, \beta, \delta, \gamma$. The results (denoted “ABC-SMC-RF(A)”) establish yet more accurate parameter means and 30 – 48% reduction in standard deviations, compared to ABC-RF (Fig. 1D). The population dynamics, in addition to maintaining similar means as the data, also exhibit significantly narrower confidence intervals. Together, the results demonstrate the strengths of ABC-SMC-RF, which combines the power of random forest approaches and the ABC-SMC framework’s capacity to refine particle proposals.

Finally, we integrate the automatic “Beaumont perturbations” (1) in ABC-SMC-RF(B), with other settings remaining the same as ABC-SMC-RF(A). While the mean parameters and population dynamics are similarly faithful to the true values and data observations, the confidence intervals are further concentrated, with an additional 30 – 46% reduction in parameter standard deviations compared to ABC-SMC-RF(A)’s posterior distributions. This establishes that similar to the traditional ABC-SMC setting, the “Beaumont perturbations” remain remarkably efficient in ABC-SMC-RF applications.

B. Inferring mutation clusters and neutral tails in the Site Frequency Spectrum

Site Frequency Spectrum (SFS) is a common and powerful statistic to analyze mutations and reconstruct cancer clonality from DNA-sequencing data. For each detected mutation i , the genomic data consists of (z_i, r_i) , where z_i is the number of sequenced DNA fragments that contain the mutation (“variant reads”) among a total of r_i fragments sequenced at the genomic location (“total reads”). The mutation’s variant allele frequency (VAF) is then defined as $f_i = z_i/r_i$. The SFS counts the number of mutations whose VAFs fall within each bin $(f_1, f_2]$ across the full interval $[0, 1]$ (Fig. 2A).

The mutations that a cancer sample’s most recent common ancestor (MRCA) harbors are present in all cancer cells. Assuming there are two alleles of each genomic site (e.g., in a normal human genome), these mutations are expected to

cluster at $\text{VAF} \approx \rho/2$, where ρ is the proportion of cancer cells in the sequenced sample (orange mutations, Fig. 2A). If there are ongoing expansions of cancer cell subsets, the mutations in the MRCA of each subclone h produce another cluster in the SFS at $\text{VAF} \approx \rho \cdot c_h/2$, where c_h is the proportion of subclone h among the cancer cells (blue mutations, Fig. 2A). In addition, cells across all subclones in the sample accumulate further mutations as they divide (gray mutations, Fig. 2A). From previous work in population genetics, we expect these mutations to form a “neutral tail” in the SFS that follows

$$\text{Mutation count} \propto \text{VAF}^{1/\alpha}$$

where the power α reflects the ongoing tumor expansion rate.

Among ≥ 30 currently available mutation clustering algorithms, only one method seeks to detect the SFS tail [13] despite theoretical predictions of its existence in the last several decades, likely because of several practical factors. First, it can be difficult to distinguish the peak of the neutral tail from a rare subclone (i.e., with low VAF). Second, the tail often overlaps significantly with multiple mutation clusters. Therefore, careful consideration is required to ascertain that the improvement in SFS fit by including the tail is significant enough to offset the associated increase in model entropy. Finally, the often low DNA-sequencing depth and high false-negative rates of current mutation calling pipelines in combination result in the exclusion of most mutations with low VAFs in the data, which make up most of the neutral tail.

We have previously established the mathematical framework to correct the SFS formulation for the effects of low DNA-sequencing coverage and mutation filtering [14]. We assume a cancer sample with H subclones, where subclone h makes up a fraction c_h of the sample. The expected number of mutations, where each is present in k copies among all n available fragments in the sample, is then

$$\mathbb{E}(S(k)|\theta) = \omega_0 \cdot g_{\text{tail}}(k|\alpha) + \sum_{h=1}^H \omega_h \cdot g_{\text{peak}}(k|p_h) \quad (3)$$

where the ω_h mutations in cluster h follow distribution $g_{\text{peak}}(k|p_h)$, which is binomial with n trials and success rate $p_h = c_h/2$, and the ω_0 mutations in the tail follow distribution $g_{\text{tail}}(k|\alpha) \propto 1/k^\alpha$ for $k = 1, \dots, n \cdot \max(p_1, \dots, p_H)$.

Eq. (3) provides the exact SFS, when all DNA fragments are sequenced. To account for sequencing effects, we define $\varphi(r)$ to be the probability that a genomic site is sequenced with a total of r reads. This probability distribution is empirically available by tallying the total reads $\{r_i\}$ in the data. In [14], we showed that the expected number of mutations i with observed VAF $f_i = z_i/r_i$ in a given bin $(f_1, f_2] \subset [0, 1]$ is

$$\mathbb{E}(S((f_1, f_2])|\theta) = \sum_{k=1}^n \mathbb{E}(S(k)|\theta) \sum_r \varphi(r) \times \mathbb{P}\left(\text{Binomial}\left(r, \frac{k}{n}\right) \in (f_1 r, f_2 r]\right) \quad (4)$$

In addition to the coverage corrections, the effects of biased mutation calling can also be mitigated by imposing read count

TABLE I
COMPARISON OF POSTERIOR MEANS $\pm$ STANDARD DEVIATIONS FOR THE
NEUTRAL TAIL AND MUTATION CLUSTER PARAMETERS IN SAMPLE
TCGA-AA-3977 FROM ABC-RF, ABC-SMC-RF(A) AND (B)

	ABC-RF	ABC-SMC-RF(A)	ABC-SMC-RF(B)
α	1.85 ± 0.51	1.15 ± 0.08	1.39 ± 0.01
p_1	0.43 ± 0.13	$0.38 \pm 9.2 \cdot 10^{-3}$	$0.36 \pm 4.9 \cdot 10^{-5}$
p_2	0.23 ± 0.07	0.31 ± 0.03	$0.25 \pm 5.8 \cdot 10^{-4}$
$\omega_0^{(*)}$	371692 ± 169821	613607 ± 50300	517544 ± 1429
$\omega_1^{(*)}$	305282 ± 223842	294598 ± 95399	401859 ± 207
$\omega_2^{(*)}$	320440 ± 220600	166464 ± 70824	147119 ± 1963

thresholds (L, M) , where mutations with total read count $r_i < M$ or variant read count $z_i < L$ in the data are filtered out. The same conditioning can be imposed in our SFS formula by replacing $\varphi(r)$ with the truncated coverage distribution

$$\varphi_{L,M}(r) = \begin{cases} \frac{|i: z_i \geq L, r_i = r|}{|i: z_i \geq L, r_i \geq M|}, & \text{if } r \geq M \\ 0, & \text{otherwise} \end{cases}$$

and the binomial probability in Eq. (4) with the corrected probability $\mathbb{P}(\text{Binomial}(r, \frac{k}{n}) \in (f_1 r, f_2 r] \cap [L, \infty))$. Eq. (4) is then directly comparable to the data after filtration.

The parameters necessary to fit our SFS formulation to the data consist of $\alpha, p_1, \dots, p_H, \omega_0, \dots, \omega_H$. However, we have previously observed that the tail parameters are particularly sensitive: a slight increase in tail power α necessitates a significantly higher mutation count ω_0 to make up the same number of observed mutations in Eq. (4). Thus, instead we seek to infer $\omega_0^{(*)}, \dots, \omega_H^{(*)}$, where $\omega_h^{(*)}$ is the number of observed mutations in each component h . Conversion between ω_h and $\omega_h^{(*)}$ is straightforward, as $\omega_h^{(*)} = \mathbb{E}(S((0, 1])|\theta)$ where $\mathbb{E}(S(k)|\theta)$ is replaced with $\omega_h \cdot g_{\text{tail}}(k|\alpha)$ for $h = 0$ and $\omega_h \cdot g_{\text{peak}}(k|p_h)$ for $h = 1, \dots, H$. We thus aim to infer the converted parameter set $\theta = \{\alpha, p_1, \dots, p_H, \omega_0^{(*)}, \dots, \omega_H^{(*)}\}$.

We examine the application of our SFS framework to analyze whole-genome sequencing data for sample AA-3977 from The Cancer Genome Atlas (TCGA) [14], from a patient with colon cancer displaying Polymerase ϵ mutator phenotype. We seek fits with a tail and $H = 2$ clusters, with priors

$$\begin{aligned} \pi(\alpha) &\sim \text{Uniform}(0.9, 5) \\ \pi(p_1), \pi(p_2) &\sim \text{Uniform}(0.01, 1) \\ \pi(\omega_0^{(*)}), \pi(\omega_1^{(*)}), \pi(\omega_2^{(*)}) &\sim \text{Uniform}(0, 2 \cdot N) \end{aligned}$$

where N is the number of observed mutations after filtration.

We compare three methods to infer the posterior distributions for $\{\alpha, p_1, p_2, \omega_0^{(*)}, \omega_1^{(*)}, \omega_2^{(*)}\}$. ABC-RF is implemented with 10000 particles. Also totaling 10000 particles, ABC-SMC-RF(A) instead runs 10 ABC-SMC-RF iterations, each with 1000 particles and heuristic parameter perturbations

$$\begin{aligned} K(\alpha|\alpha') &= \alpha' + \mathcal{N}(0, \sigma^2 = 0.3^2) \\ K(p_h|p'_h) &= p'_h + \mathcal{N}(0, \sigma^2 = 0.1^2) \\ K(\omega_h^{(*)}|\omega'_h) &= \omega'_h + \mathcal{N}(0, \sigma^2 = (0.1 \cdot N)^2) \end{aligned}$$

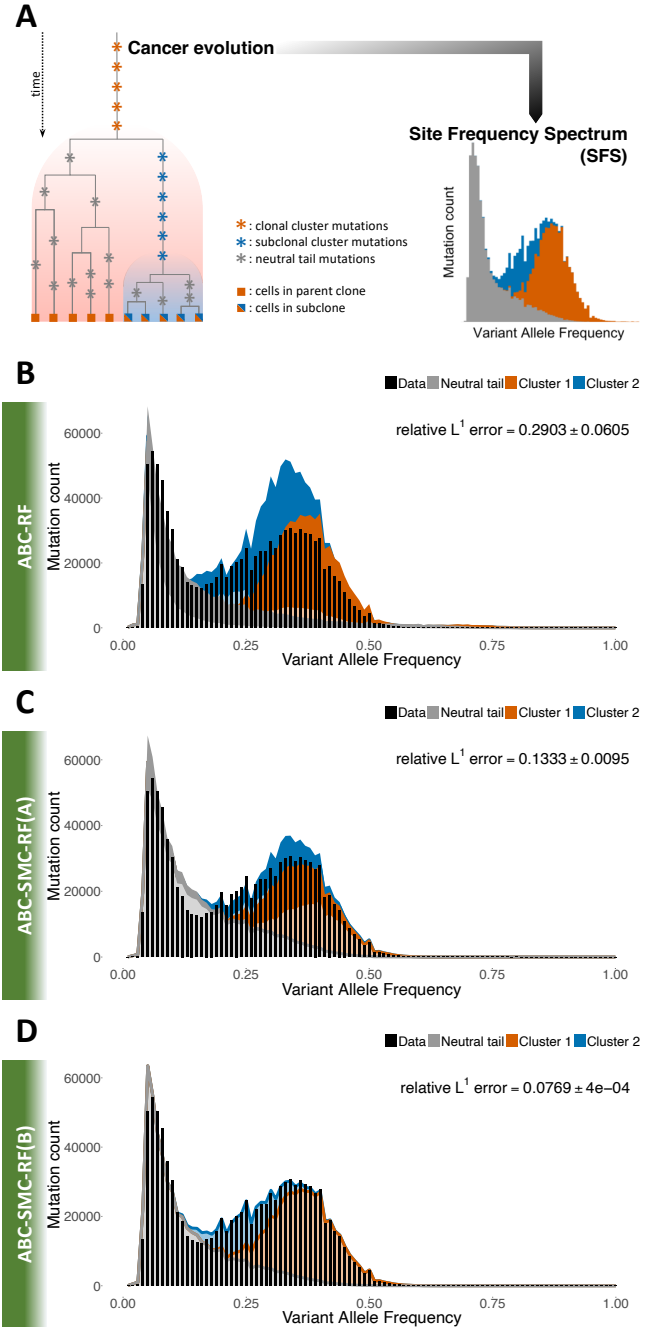


Fig. 2. **A:** Illustration of the connection between the Site Frequency Spectrum (SFS) from a cancer sample and the clonal evolution preceding it. Clonal mutations (orange stars) occur before the tumor's most recent common ancestor (MRCA) and are present in all cancer cells (orange squares), thus forming a cluster in the SFS at the highest mean Variant Allele Frequency (VAF). Subclonal mutations (blue stars) occur before each subclone's MRCA and are found in its progeny (orange/blue squares), creating an SFS cluster at lower VAFs. Other mutations (gray stars) mainly result from recent mitoses among every cell (all squares), resulting in a "neutral tail" that peaks in the low-VAF region. **B-D:** Comparison of the SFS from diploid mutations in sample TCGA-AA-3977 (histograms) against the expected observed SFS with neutral tail (gray regions) and 2 clusters (orange and blue regions), computed with parameters inferred from ABC-RF (**B**), ABC-SMC-RF(A) (**C**) and ABC-SMC-RF(B) (**D**). The width of each darkly shaded region indicates the mean $\pm$ standard deviation of the corresponding neutral tail or cluster component.

while ABC-SMC-RF(B) utilizes the same numbers of iterations and particles, but with “Beaumont perturbations”. The statistic associated with each particle is the relative L^1 error between the observed SFS and the expectations from Eq. (4), both computed with 100 bins of size 0.01 and filtered with $(L, M) = (3, 42)$. Because the mutation cluster order does not impact the expected SFS (i.e., switching $(p_1, \omega_1^{(*)})$ and $(p_2, \omega_2^{(*)})$ results in the same fit), we facilitate fair comparison and accurate parameter convergence by reordering clusters in each particle such that $p_1 \geq p_2$ in the ABC-RF posterior and at the end of each ABC-SMC-RF iteration.

The ABC-RF posteriors remain widespread, indicating incomplete convergence (Table I). In contrast, both ABC-SMC-RF variants achieve substantially tighter parameter estimates, indicating increased confidence despite utilizing the same number of particles. In particular, ABC-SMC-RF(B) produces the most precise estimates for all parameters. Notably, ABC-SMC-RF(B)’s precision is accompanied by elevated accuracy of the simulated SFS compared to the data (Fig. 2D), with 73% and 42% improvements in relative L^1 errors compared to ABC-RF and ABC-SMC-RF(B), respectively (Fig. 2B-C).

These results suggest that the integration of our ABC-SMC-RF inference method, automatic “Beaumont perturbations” and our SFS modeling framework is able to provide a robust and reliable approach to infer cancer subclones and detect ongoing tumor expansion from DNA-sequencing data. As treatment outcome and whether cancers relapse or metastasize can often be linked to both subclonal heterogeneity and tumor growth rate across different cancer subtypes, this future initiative holds the potential to offer substantial help toward improving personalized treatment design in the clinic.

VI. CONCLUSION

The adoption of random forests in Approximate Bayesian Computation has introduced a novel approach to likelihood-free inference. In developing ABC-SMC-RF, we integrated the robustness of these methods to noise and user-defined hyper-parameters with the SMC framework’s ability to refine posterior distributions iteratively, toward improving both accuracy and efficiency. However, ABC-SMC-RF’s requirement of parameter perturbation strategies remains a consequential decision from the user that can impact the results considerably.

In this paper, we explore the application of the adaptive Gaussian perturbations for ABC-SMC from [8] in the context of ABC-SMC-RF. Although both rooted in biology and containing multiple parameters, the applications we consider are wide-ranging: the Lotka-Volterra model in Section V-A is stochastic, ecological and high-dimensional, whereas our SFS model in Section V-B is deterministic, focused on genomics and can be summarized by a single error term. However, the observations remain consistent. In both cases, ABC-SMC-RF yields results that are both more accurate and more precise than previous ABC approaches, both with and without random forests. Furthermore, in addition to eliminating the typical trials-and-errors necessary to select parameter pertur-

bation schemes, the automatic scheme developed in [8] leads to optimized ABC-SMC-RF results compared to heuristic choices. We thus have incorporated this strategy as the default perturbation kernels in ABC-SMC-RF, publically available at <https://github.com/dinhngockhanh/abcsmc-rf>. We expect that this will transform ABC-SMC-RF into a convenient and effective parameter inference method for problems in both biology and other sciences.

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GENERATIVE AI DISCLOSURE STATEMENT

During the preparation of this work, the authors used ChatGPT in order to improve the language and clarity of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

REFERENCES

- [1] Tavaré, S., Balding, D. J., Griffiths, R. C. and Donnelly, P. *Inferring coalescence times from DNA sequence data*, Genetics 145(2):505-18, 1997 <https://doi.org/10.1093/genetics/145.2.505>
- [2] Marjoram, P., Molitor, J., Plagnol, V. and Tavaré, S. *Markov chain Monte Carlo without likelihoods*, Proceedings of the National Academy of Sciences 100(26):15324-8, 2003 <https://doi.org/10.1073/pnas.0306899100>
- [3] Toni, T., Welch, D., Strelkowa, N., Ipsen, A. and Stumpf, M. P. *Approximate Bayesian computation scheme for parameter inference and model selection in dynamical systems*, Journal of the Royal Society Interface 6(31):187-202, 2009 <https://doi.org/10.1098/rsif.2008.0172>
- [4] Raynal, L., Marin, J.-M., Pudlo, P., Ribatet, M., Robert, C. P. and Estoup, A. *ABC random forests for Bayesian parameter inference*, Bioinformatics 35(10):1720-8, 2019 <https://doi.org/10.1093/bioinformatics/bty867>
- [5] Čevič, D., Michel, L., Näf, J., Bühlmann, P. and Meinshausen, N. *Distributional random forests: Heterogeneity adjustment and multivariate distributional regression*, Journal of Machine Learning Research 23(333):1-79, 2022 <https://doi.org/10.3929/ethz-b-000585771>
- [6] Dinh, K. N., Liu, C., Xiang, Z., Liu, Z. and Tavaré, S. *Approximate Bayesian computation sequential Monte Carlo via random forests*, Statistics and Computing 35(6):219, 2025 <https://doi.org/10.1007/s11222-025-10748-x>
- [7] Sisson, S. A., Fan, Y. and Beaumont, M. *Handbook of approximate Bayesian computation*, CRC Press, 2018 <https://doi.org/10.1201/9781315117195>
- [8] Beaumont, M., Cornuet, J.-M., Marin, J.-M. and Robert, C. P. *Adaptive approximate Bayesian computation*, Biometrika 96(4):983-90, 2009 <https://doi.org/10.1093/biomet/asr052>
- [9] Lotka, A. J. *Elements of physical biology*, Williams & Wilkins, 1925
- [10] Volterra, V. *Variazioni e fluttuazioni del numero d’individui in specie animali conviventi*, Società anonima tipografica “Leonardo da Vinci”, 1926
- [11] Gillespie, D. T. *Exact stochastic simulation of coupled chemical reactions*, The journal of physical chemistry 81(25):2340-61, 1977
- [12] Jabot, F., Faure T. and Dumoulin, N. *Easy ABC: performing efficient approximate Bayesian computation sampling schemes using R*, Methods in Ecology and Evolution 4(7):684-7, 2013 <https://doi.org/10.1111/2041-210X.12050>
- [13] Caravagna, G., Heide, T., Williams, M. J., Zapata, L., Nichol, D. and Chkhaidze, K. et al. *Subclonal reconstruction of tumors by using machine learning and population genetics*, Nature genetics 52(9):898-907, 2020 <https://doi.org/10.1038/s41588-020-0675-5>
- [14] Dinh, K. N., Jaksik R., Kimmel M., Lambert, A. and Tavaré, S. *Statistical inference for the evolutionary history of cancer genomes*, Statistical science 35(1):129, 2020 <https://doi.org/10.1214/19-sts7561>